

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                  |   | Marks available |     |     |       |       |      |
|----------|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                  |   | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 8        | (a) | (i)  |  | as pH increases, citric acid content decreases and sugar content increases                                                                                                                                                                                                                                                                                       | ✓ |                 |     | 2   | 2     | 2     |      |
|          |     |      |  | as acidity decreases, ascorbic acid content decreases and water content decreases                                                                                                                                                                                                                                                                                |   |                 |     |     |       |       |      |
|          |     |      |  | tomatoes are a good source of vitamin C and citric acid                                                                                                                                                                                                                                                                                                          |   |                 |     |     |       |       |      |
|          |     |      |  | citrus fruits contain ascorbic acid and a natural preservative                                                                                                                                                                                                                                                                                                   | ✓ |                 |     |     |       |       |      |
|          |     | (ii) |  | award (1) for <b>any</b> of following <ul style="list-style-type: none"> <li>the values for ascorbic acid are bigger but the unit is 1000 times smaller</li> <li>mg/100g is a smaller unit than %</li> <li>citric acid is measured in % but ascorbic acid is measured in mg/100g</li> <li>there is much more citric acid (than ascorbic acid) present</li> </ul> |   |                 |     | 1   | 1     | 1     |      |
|          | (b) | (i)  |  | $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$<br>formulae (1)<br>state symbols (1)                                                                                                                                                                                                                                   |   | 2               |     |     | 2     |       |      |